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$$f(x,y,z) = \sqrt{xyz} \text{ in } a(2,-1,-2) \text{ richting } h(1,2,-2) \text{ en } f(2,-1,-2) = 2$$

$$Df(a,h) = \lim_{\lambda \to 0} \frac{f\left((2,-1,-2) + \lambda(1,2,-2)\right) - 2}{\lambda}$$
$$= \lim_{\lambda \to 0} \frac{\sqrt{-2\lambda^3 - 4\lambda^2 + 2\lambda + 4} - 2}{\lambda}$$
$$= \lim_{\lambda \to 0} \frac{\sqrt{-2\lambda^3 - 4\lambda^2 + 2\lambda + 4} - 2}{\lambda} \cdot \frac{\sqrt{-2\lambda^3 - 4\lambda^2 + 2\lambda + 4} + 2}{\sqrt{-2\lambda^3 - 4\lambda^2 + 2\lambda + 4} + 2}$$
$$= \lim_{\lambda \to 0} \frac{-2\lambda^3 - 4\lambda^2 + 2\lambda + 4 - 4}{\lambda \cdot \left(\sqrt{-2\lambda^2 - 4\lambda^2 + 2\lambda + 4} + 2\right)}$$
$$= \lim_{\lambda \to 0} \frac{\lambda\left(-2\lambda^2 - 4\lambda + 2\right)}{\lambda\sqrt{-2\lambda^3 - 4\lambda^2 + 2\lambda + 4} + 2} = -\frac{1}{2}$$

References